



An innovative thermal management control strategy for electric scooters to enhance performance and efficiency

Chun-Hsin Chang¹ · Guan-Xun Wang² · Hwa-Dong Liu³

Received: 26 April 2025 / Accepted: 12 June 2026 / Published online: 12 July 2026
© The Author(s) 2026

Abstract

This study proposes a temperature-dependent control strategy for improving the thermal management performance and energy efficiency of electric scooters under varying ambient conditions. A heat-exchange-based experimental platform was developed to replicate real vehicle operation for a 6.4-kW electric scooter, with ambient temperatures controlled between 30 and 50 °C. The proposed strategy dynamically adjusts the water pump duty cycle based on real-time coolant temperature, which reflects the thermal states of key components, including the traction motor and the motor control unit (MCU), eliminating the need for complex optimization algorithms. Experimental validation was conducted under both the WMTC Part 1 driving cycle (2400 s) and a constant-speed condition of 50 km h⁻¹ (1100 s). The results demonstrate significant energy-saving performance. Under the WMTC Part 1 cycle, the proposed strategy reduced average pump power consumption from 18.35 W to as low as 0.94 W at 30 °C, achieving a maximum power reduction of 17.41 W and an energy improvement rate of 94.86%. Similarly, under constant-speed operation, the maximum power reduction reached 16.38 W, with an energy improvement rate of 89.68%. Across all tested conditions, the coolant temperature—representing the thermal behavior of the motor and MCU—remained well controlled, with deviations within approximately 0.17–1.59 °C, indicating effective thermal stability. These findings confirm that the proposed rule-based control strategy can significantly reduce energy consumption while maintaining adequate thermal protection for critical components. The approach offers a practical and low-complexity solution that can be readily implemented in vehicle control units (VCUs) or embedded systems. This makes it particularly suitable for small electric vehicles operating in high-temperature environments, with potential benefits for improving overall energy efficiency and extending driving range.

Keywords Electric scooter · Thermal exchange cooling platform · Proposed optimized control strategy · Coolant temperature

Introduction

With the escalating impacts of global warming and the growing demand for sustainable transportation, electric vehicles (EVs) have emerged as a critical solution for reducing carbon emissions and decreasing reliance on fossil fuels. Governments worldwide are actively promoting EV-related policies and investing in the development of electric mobility technologies [1]–[5].

However, the performance and reliability of EVs are significantly influenced by environmental conditions, particularly ambient temperature. Previous studies have demonstrated that both high and low temperatures can substantially affect the driving range and efficiency of EVs. For instance, Hamwi et al. [6] reported that at low temperatures (10 °C), the driving range of EVs can decrease by approximately

✉ Hwa-Dong Liu
hdliu@ntnu.edu.tw
Chun-Hsin Chang
cchsin@ntnu.edu.tw
Guan-Xun Wang
61170023H@gapps.ntnu.edu.tw

¹ Institute and Undergraduate Program of Vehicle and Energy Engineering, National Taiwan Normal University, Taipei 106, Taiwan

² Department of Industrial Education, National Taiwan Normal University, Taipei 106, Taiwan

³ Undergraduate Program of Vehicle and Energy Engineering, National Taiwan Normal University, Taipei 106, Taiwan

25%, while at high temperatures (around 40 °C), the range may decline by about 22%. Similarly, Yuksel et al. [7] showed that regional temperature variations can result in range reductions of up to 36%. These findings highlight that thermal effects play a crucial role in determining EV performance under real-world operating conditions.

In addition to impacting driving range, elevated temperatures can lead to severe thermal management challenges in key components such as electric motors and batteries. For electric motors, overheating may result in demagnetization, degradation of insulation performance, reduced efficiency, and even system failure [8]–[10]. To mitigate these risks, various cooling strategies have been investigated, including air-cooling, liquid-cooling, and hybrid-cooling systems [11], [12]. For example, Tian et al. [13] proposed an integrated thermal management system capable of increasing the driving range by up to 31.71%. Furthermore, the study by Szumska et al. [14] indicates that temperature variations significantly influence energy consumption under different environmental conditions.

For battery systems, temperature regulation is even more critical. Elevated temperatures can accelerate battery degradation, reduce capacity, and increase safety risks such as thermal runaway [15]–[16]. Wang et al. [17] demonstrated that advanced cooling strategies can effectively reduce battery temperature; however, their effectiveness remains limited under extreme environmental conditions. Similarly, Ouyang et al. [18] reported that temperature control has a significant impact on battery performance in harsh environments. The application of phase change materials, as discussed in [19], further highlights the importance of efficient thermal management solutions for enhancing thermal stability.

Although extensive research has been conducted on EV thermal management systems, most existing studies focus on fixed cooling strategies or simulation-based approaches. Comparatively little attention has been given to real-time adaptive control strategies that can simultaneously balance thermal performance and energy efficiency, particularly for electric scooters operating under varying ambient temperatures.

Existing studies indicate that thermal management is essential for improving the safety, efficiency, and driving range of EVs. However, the majority of published research primarily concentrates on battery thermal management, simulation-based analyses, or fixed cooling strategies designed for general EV platforms. In contrast, practical thermal control strategies for electric scooters—especially those capable of operating under dynamic environmental temperatures and real-time conditions—remain underexplored. Moreover, although some studies have investigated advanced thermal management approaches, such as integrated cooling systems, phase change materials, and optimized thermal architectures,

their practical implementation often requires complex system integration or incurs high computational costs. For small-scale electric vehicles, there is still a lack of simple, adaptive, and energy-efficient control strategies.

Therefore, building upon existing literature on EV thermal management, cooling control strategies, and energy efficiency analysis, this study addresses the following key research question: how to develop a practical and energy-efficient thermal management control strategy capable of effectively regulating coolant temperature under varying ambient conditions while minimizing energy consumption. To address this challenge, this study proposes a thermal management control strategy for a 6.4-kW electric scooter based on a heat exchange cooling platform. The proposed approach dynamically adjusts the water pump duty cycle according to the coolant temperature, thereby achieving adaptive cooling performance.

The main contributions of this study are summarized as follows:

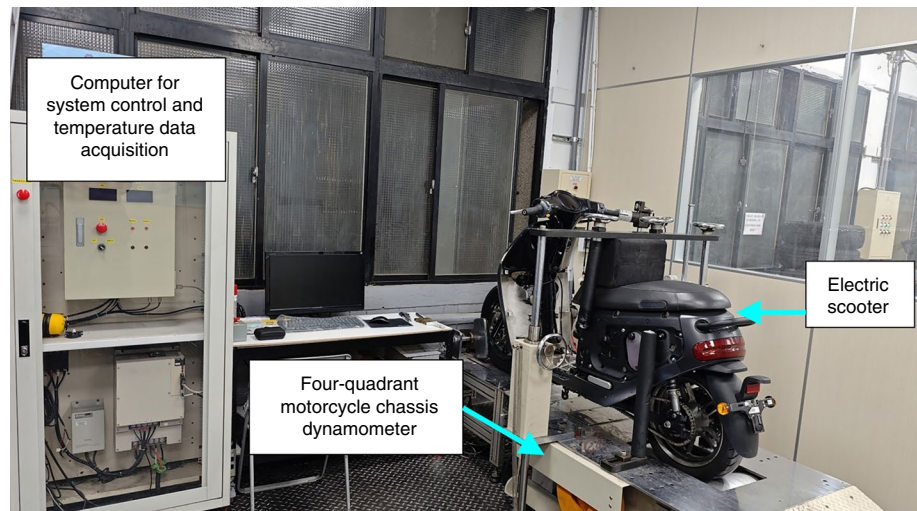
- (i) A heat-exchange-based cooling simulation platform is developed, capable of reproducing real vehicle operating conditions under controlled ambient temperatures.
- (ii) A temperature-dependent control strategy is proposed, which adaptively regulates the duty cycle of the water pump.
- (iii) Experimental validation is conducted under both the WMTC driving cycle and constant-speed conditions. The results demonstrate that the proposed strategy can significantly reduce energy consumption while maintaining thermal stability.
- (iv) A practical and implementable solution for thermal management systems in real-world electric scooters is provided.

Experimental design

Thermal behavior sampling of the cooling system

The electric scooter used in this study is manufactured by Gogoro, a brand with over 200,000 units sold in Taiwan. The motor has a rated power of 6.4 kW. The vehicle is mounted on a four-quadrant motorcycle chassis dynamometer to capture the thermal behavior and temperature evolution trends, as shown in Fig. 1.

The primary objective is to obtain the coolant temperature profile of the liquid-cooling system under realistic operating conditions. This temperature trend is then reproduced on a simulation platform, which serves as the experimental platform for subsequent analyses in this study. This approach reduces data uncertainty associated with repeated

Fig. 1 Four-quadrant motorcycle chassis dynamometer

dynamometer testing and improves the consistency of experimental results.

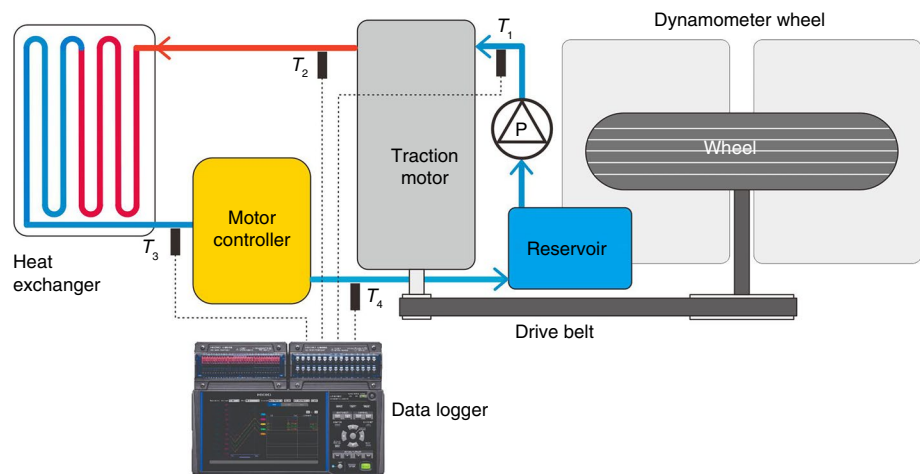
As shown in Fig. 2, the cooling system of the electric scooter is illustrated. Type-T thermocouples (Copper–Constantan) are installed at specific locations to monitor temperature: at the water pump outlet/motor inlet (T_1), motor outlet/heat exchanger inlet (T_2), heat exchanger outlet/motor control unit (MCU) inlet (T_3), and MCU outlet/reservoir inlet (T_4). The sensing junctions of all four thermocouples are in direct contact with the coolant to ensure measurement accuracy. The acquired temperature data are recorded using a data logger.

The electric motor transmits power to the rear wheel via a belt drive, and the rear wheel, in turn, drives the roller of the dynamometer. The resistance generated by the roller is used to simulate road load. Neglecting road gradients and external wind effects (headwind or tailwind), a four-quadrant motorcycle chassis dynamometer is employed to emulate road resistance. The total road load model is expressed as follows:

$$F = mgf_r + \frac{1}{2}\rho A_f C_D + m_\delta \frac{dv}{dt} \quad (1)$$

where m is the vehicle mass, g is the gravitational acceleration, f_r is the rolling resistance coefficient, ρ is the air density, A_f is the frontal area, C_D is the aerodynamic drag coefficient, m_δ is the rotational mass factor, and dv/dt represents the vehicle acceleration.

After the total road load is defined, the experiments are conducted. To ensure realistic operating conditions, all experiments are performed in a laboratory with controllable ambient temperatures ranging from 20 to 50 °C. In this study, two driving profiles are adopted: WMTC Part 1 and a custom constant-speed condition of 50 km h⁻¹. The WMTC Part 1 test is initially conducted for 600 s; however, to ensure data stability, the experiment is extended to 2400 s. Similarly, under the constant-speed condition of 50 km h⁻¹, the test duration is set to 1100 s to achieve stable experimental data. Finally, the temperature profiles obtained under these two driving conditions are recorded and used as the basis for

Fig. 2 Experimental setup of the electric scooter on the dynamometer and temperature measurement configuration

defining the heat source in the development of the thermal simulation platform.

Experimental platform for heat exchange and cooling system

Figure 3 shows the schematic architecture of the heat exchange cooling system simulation platform, while Fig. 4 illustrates the physical simulation platform constructed in

this study. Table 1 presents the specifications of the equipment used in the platform. The heating system in this study was designed based on power data obtained from actual vehicle measurements. Stainless steel immersion-type electric heating tubes were employed to simulate the heat sources generated by the motor and the MCU. In the water circulation system of the simulation platform, T-type temperature sensors were installed, and coolant temperature data at various pipeline sections were collected using

Fig. 3 Schematic architecture of the heat exchange cooling system simulation platform

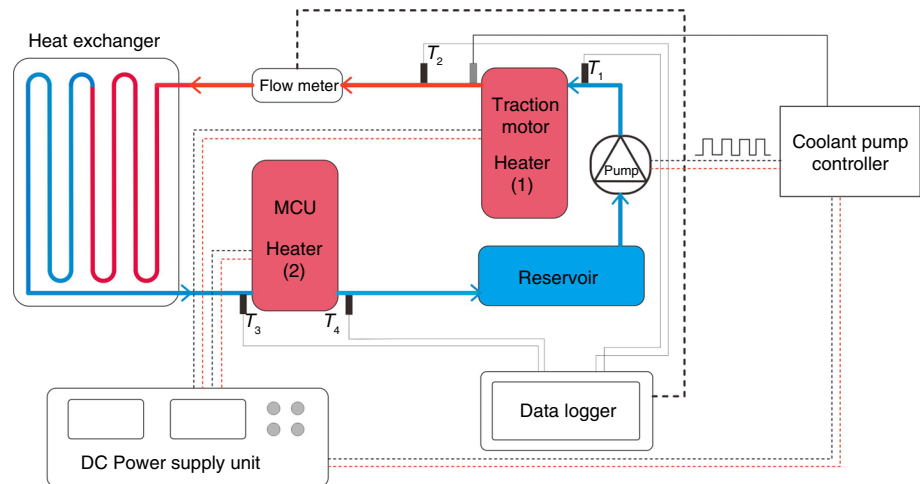


Fig. 4 Physical simulation platform for heat exchanger-based cooling system

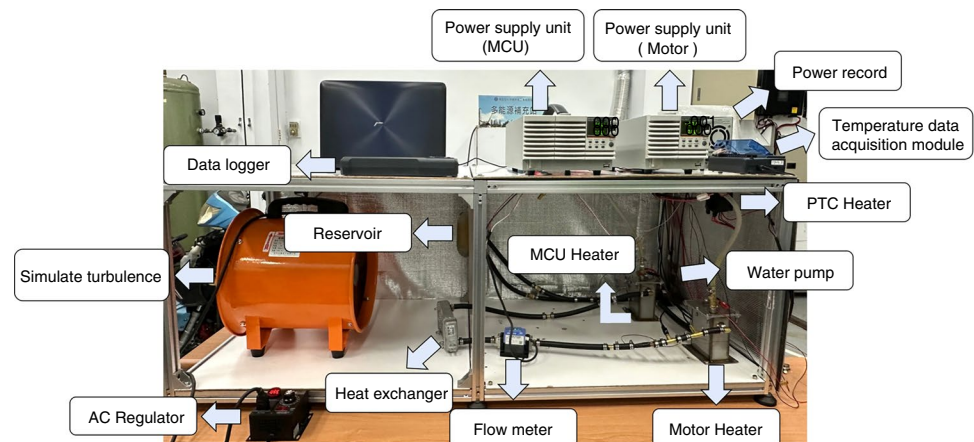


Table 1 Equipment specifications of the physical simulation platform for the heat exchange cooling system

Equipment	Specification
DC Power supply unit	Brand: GW Instek, Model: PSW 80–27
Temperature data acquisition device	Brand: HIOKI, Model: LR8450-01
Controller	Brand: WAVGAT, Model: ZS-X11F
Flow meter	Brand: Regal Joint, Model: KSL
Electric heating tube	Brand: Qing Da Electric Heating Co., Specification: U-shape, 70 V/700W
Heat exchanger	Brand: Uni-Calsonic, Model: BX2RAD
Positive temperature coefficient heater	Brand: Maker, Specification: Insulated type, 24 V/400W
Portable cooling fan	Brand: LeGou, Specification: 12-inch fan, 110 V/2.33A

a data acquisition system. The measurement points were located at: T1 (from the reservoir to the water pump), T2 (from the traction motor to the heat exchanger), T3 (from the heat exchanger to the MCU heater), and T4 (from the MCU heater back to the reservoir).

Due to the compact structure of the electric scooter cooling loop, the piping lengths between key components are relatively short. Therefore, temperature variations along these short connecting sections can be considered negligible. Specifically, measurement point T2 represents the coolant temperature at the outlet of the traction motor. Owing to the continuous flow path and negligible heat loss along the short pipeline, this temperature can be regarded as equivalent to the inlet temperature of the heat exchanger. Similarly, the other measurement points (T1, T3, and T4) are assumed to represent the local coolant temperatures at their respective locations, with minimal deviation along the connecting segments.

The electric heating tubes were driven by a programmable DC power supply unit (see Table 1) to simulate the thermal behavior of the traction motor and the MCU. Specifically, the power input was dynamically controlled according to real vehicle operating conditions, including the WMTC PART 1 driving cycle and a constant-speed condition at 50 km h^{-1} . This approach enables the simulation of realistic transient thermal loads under varying operating scenarios.

The cooling system utilized the original radiator of the Gogoro 2, with an industrial-grade cooling fan installed in front of the radiator to replicate the headwind cooling effect experienced during actual vehicle operation.

For water pump control, a motor signal controller was used to implement the proposed optimized control strategy. Temperature signals were received from the temperature sensors and processed according to predefined logic to generate control outputs. Currently, the water pump is operated using a temperature-based pulse width modulation (PWM)

signal, enabling the system to achieve a cooling performance equivalent to 100% water pump operation while effectively reducing energy consumption from prolonged water pump usage.

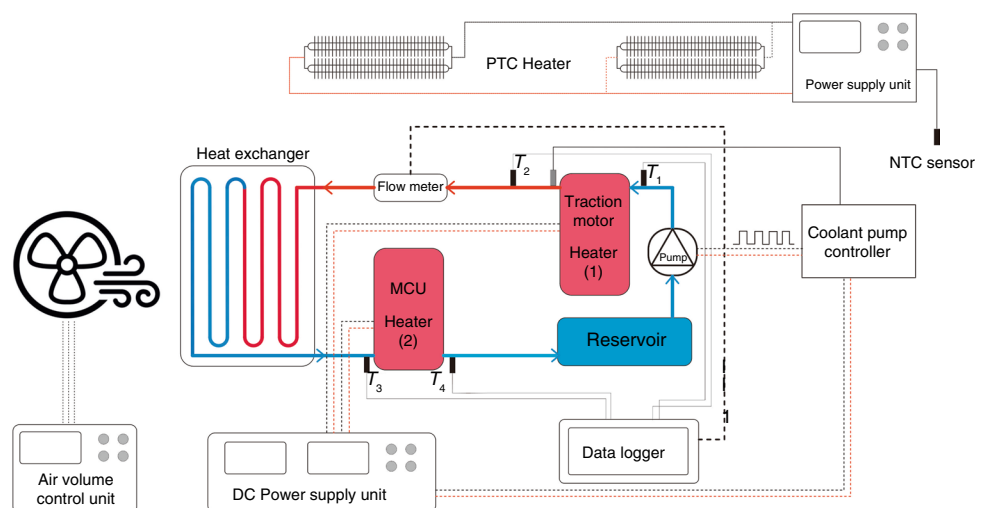
A heat exchange cooling system test platform with ambient temperature simulation capability

In response to the impact of extreme climate conditions caused by global warming, ambient temperatures have been rising year by year. The heat exchange cooling system proposed in this study (as shown in Fig. 5) is capable of regulating and simulating ambient temperature conditions, allowing for the reproduction and analysis of thermal rise phenomena that occur when EVs operate under extreme climates. Through the design and implementation of control strategies, the system can effectively maintain coolant temperature within the optimal operating range.

This study employs a previously developed heat exchange platform that is consistent with the real vehicle configuration (as shown in Fig. 5). Two sets of Positive Temperature Coefficient (PTC) heaters are installed inside the platform, and a programmable power supply is used to elevate and control the ambient temperature. In addition, an upunuphū aluminum foil insulation layer is applied around the platform to ensure thermal stability and maintain the required experimental conditions.

A cooling fan is used to provide a steady airflow to facilitate heat dissipation. Although the airflow rate and velocity are not explicitly measured, the fan operates under a fixed setting throughout all experiments to ensure consistent thermal conditions within the test chamber. Since all experiments are conducted under identical airflow conditions, the effect of convective heat transfer remains constant across all test scenarios, enabling a fair comparison of different control strategies.

Fig. 5 Schematic diagram of the heat exchange cooling system proposed in this study



To ensure clarity and repeatability, the experimental procedure is described as follows:

First, prior to each test, the thermal simulation platform is initialized and stabilized at the target ambient temperature, ranging from 30 to 50 °C in 5 °C intervals (i.e., 30 °C, 35 °C, 40 °C, 45 °C, and 50 °C). The system is allowed to reach a steady-state condition to eliminate the influence of initial transients.

Next, two types of operating conditions are considered in this study: (i) the WMTC Part 1 driving cycle, and (ii) constant-speed operation at 50 km h⁻¹. These conditions are implemented using a programmable power supply to emulate real vehicle operation.

During each test, key parameters—including coolant temperatures at multiple measurement points (T1–T4), water pump duty cycle, and system power consumption—are continuously recorded. All measurements are conducted under identical sampling conditions to ensure consistency.

Furthermore, the duration of the WMTC driving cycle test is approximately 2400 s, while the constant-speed test lasts about 1100 s. For comparability, all experiments are conducted under identical operating conditions, with only the control strategy being varied.

Finally, each test is repeated at least three times to ensure result reliability. The collected data are then processed and analyzed to evaluate thermal performance and energy consumption characteristics.

Based on the above, it is evident that the experimental platform cannot fully replicate all real-world thermal conditions, such as solar radiation, varying airflow velocities, road heat feedback, and complex convective heat transfer. However, the primary objective of this study is to evaluate the effectiveness of the proposed control strategy under controlled and repeatable conditions. The use of simplified ambient temperature control helps isolate key variables and ensures the validity of comparisons between different control strategies. Although PTC heaters are employed to regulate the ambient temperature, the experimental platform effectively captures the dominant thermal behavior of the system. Consequently, the results provide meaningful insights into the relationships among coolant temperature, control strategies, and energy consumption.

Experimental uncertainty analysis

To evaluate the reliability of the experimental results, an uncertainty analysis was performed for the key measured variables. The primary sources of uncertainty include temperature measurement errors from T-type thermocouples (Copper–Constantan) and power measurement errors from the programmable DC power supply unit.

The T-type thermocouples employed in this study follow the standard Copper–Constantan configuration, which is

commonly used for accurate temperature measurement in the relevant range. Based on standard specifications, the temperature measurement uncertainty is approximately ± 0.5 °C. In addition, the power measurement uncertainty of the programmable DC power supply unit is within $\pm (0.1\% + 0.05 \text{ W})$. Considering these factors, the overall uncertainty of the experimental system is relatively small and does not significantly affect the comparative analysis or the main conclusions of this study.

The proposed optimized control strategy

To provide a physical foundation for the proposed control strategy, a simplified lumped thermal model is adopted in this study. The thermal behavior of the system can be described by the following energy balance equation:

$$C_{th} \cdot dT_c/dt = Q_{gen} - Q_{cool} \quad (2)$$

where C_{th} represents the effective thermal capacitance of the system, T_c denotes the coolant temperature, Q_{gen} is the heat generated by the motor and the motor control unit (MCU) under operating conditions, an Q_{cool} is the heat removed by the cooling system. The cooling term Q_{cool} is influenced by the coolant flow rate and heat transfer performance, both of which are directly related to the duty cycle of the water pump. Therefore, adjusting the water pump duty cycle enables effective control of the system's heat dissipation capability and the resulting temperature response.

The heat generation of the motor and MCU (Q_{gen}) is primarily associated with the operating load and electrical losses, and can be approximated as a function of power consumption during system operation. To effectively remove this heat, the performance of the cooling system largely depends on the coolant mass flow rate ($\dot{m}$), which is directly controlled by the water pump duty cycle and consequently determines the overall cooling capacity.

From a heat transfer perspective, the system's heat dissipation behavior can be expressed based on the principle of energy conservation as:

$$Q_{cool} = \dot{m} \cdot C_p \cdot (T_{out} - T_{in}) \quad (3)$$

where C_p is the specific heat capacity of the coolant, and T_{in} and T_{out} represent the inlet and outlet temperatures, respectively. This relationship reflects the direct influence of coolant flow rate and temperature difference on heat dissipation capacity.

In addition, although the heat exchanger efficiency and convective heat transfer performance are not explicitly incorporated as separate parameters in the model, their effects are implicitly embedded within the overall cooling process and can be characterized through experimentally

measured temperature responses. Therefore, even without individually resolving each heat transfer parameter, this study is able to comprehensively evaluate the thermal management performance of the proposed control strategy under practical operating conditions.

The conventionally designed 6.4 kW electric scooter compatible with the Gogoro 2 employs a water pump operating at 100% capacity to cool the internal systems. While this approach can effectively reduce temperature, it also results in higher energy consumption and may not deliver optimal thermal regulation performance.

It is assumed that the coolant remains stable and incompressible during operation. Furthermore, the thermal response of the system is assumed to be primarily governed by the coolant temperature (T_c), while minor transient fluctuations are neglected. It is also assumed that the heat generation of the motor and MCU under the specified driving cycles is consistent with the measured operating conditions.

In addition, the ambient temperature in each experimental scenario is treated as a controlled constant parameter. These assumptions are reasonable for the experimental setup used in this study and allow for a clear relationship to be established between temperature variations and control responses. Based on these assumptions, the proposed temperature-dependent control strategy is capable of effectively capturing the dominant thermal behavior of the system while maintaining practical applicability.

The objective of the proposed control strategy is to minimize the rise in coolant temperature while reducing the power consumption of the cooling system. This can be conceptually expressed as:

$$J = w_1 \cdot (T_c - T_{\text{ref}}) + w_2 \cdot P_{\text{pump}} \quad (4)$$

where T_c is the coolant temperature, T_{ref} is the reference temperature, P_{pump} denotes the pump power, and w_1 and w_2 are weighting factors.

The duty cycle settings are determined based on experimental observations of the system's thermal response. Different temperature ranges are defined to ensure sufficient cooling performance while avoiding unnecessary energy consumption.

To enhance the driving range of electric vehicles, this study proposes a novel control strategy for regulating the coolant temperature in electric scooters, thereby improving overall operational efficiency. As shown in Table 2, the operating duty cycle of the water pump is defined to maintain the coolant temperature below 60 °C. Although no formal optimization algorithm is employed, the proposed control strategy can be interpreted as a rule-based optimization approach that effectively balances thermal performance and energy efficiency.

Table 2 Coolant temperature ranges and corresponding water pump duty cycles in the proposed optimized control strategy

Coolant temperature range/°C	Water pump duty cycle, $V_{\text{pump}}/\%$
30–40	20
40–50	40
50–60	60
60–65	80
Above 65	100

When the coolant temperature falls below 30 °C, the water pump remains inactive. As the temperature rises, the water pump's duty cycle increases accordingly. Table 2 presents the temperature ranges and corresponding water pump operation cycles defined by the proposed optimized control strategy. By gradually adjusting the water pump's duty cycle, the cooling system can operate more efficiently, thereby reducing unnecessary energy consumption and extending the EV's driving range. In addition, the proposed optimized control strategy facilitates the motor in reaching its optimal operating temperature more rapidly. When the temperature exceeds the ideal range, the system activates a cooling mechanism to effectively regulate the motor temperature, maintaining it within the optimal operating range and thereby enhancing the overall operational efficiency of the EV.

The duty cycle settings in Table 2 were determined based on repeated experimental observations of the thermal response of the system under different coolant temperature ranges. For each temperature interval, the water pump duty cycle was gradually adjusted to identify a suitable operating level that could suppress temperature rise while minimizing unnecessary energy consumption. Therefore, the proposed settings represent an experimentally derived rule-based control strategy rather than the result of a formal optimization algorithm.

The data from Table 2 were used to plot the relationship between the coolant temperature range (T_c) and the water pump duty cycle (V_{pump}) using Excel, as illustrated in Fig. 6. This relationship is described using a second-order polynomial equation, as detailed below.

$$V_{\text{pump}} = a \cdot T_c^2 + b \cdot T_c + c \quad (5)$$

The parameters a , b , and c are 7×10^{-15} , 0.2, and 1×10^{-13} , respectively.

Figure 7 illustrates the flowchart of the proposed optimized control strategy. The system first measures the coolant temperature (T_c). When the temperature exceeds 30 °C, Eq. (5) is used to estimate the water pump duty cycle (V_{pump}), and the water pump is activated to assist in heat dissipation, thereby

Fig. 6 Relationship between coolant temperature range and water pump duty cycle

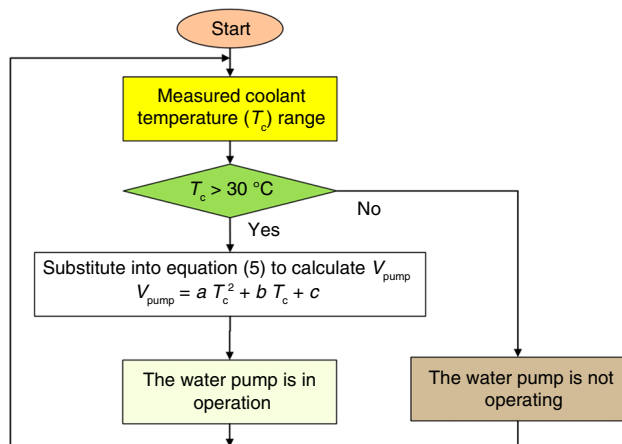
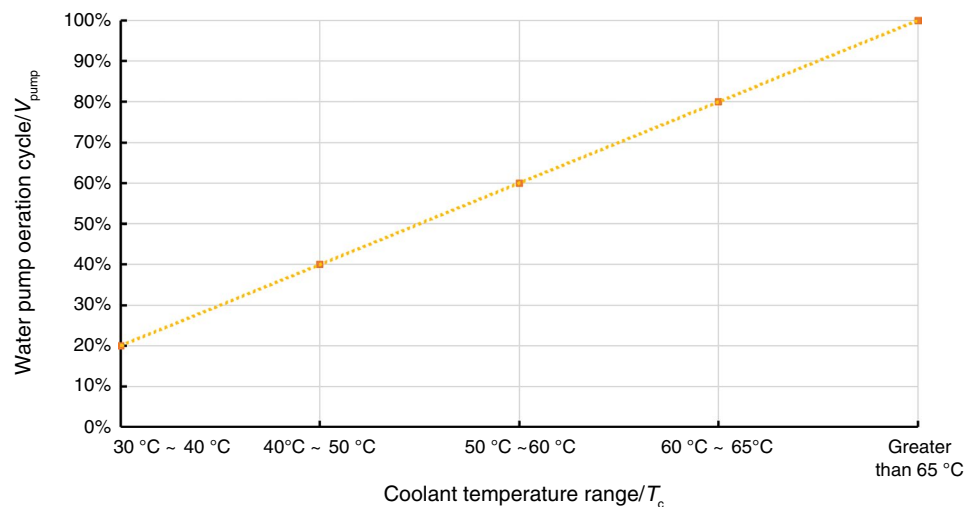


Fig. 7 Flowchart of the proposed optimized control strategy

suppressing temperature rise. Conversely, when the coolant temperature (T_c) is below 30 °C, the water pump remains inactive.

Experimental results and discussion

The experiments in this study were conducted under both the WMTC PART 1 mode and a custom constant-speed mode at 50 km h⁻¹. The evaluation focuses on power consumption characteristics, energy efficiency performance, and coolant temperature behavior. The results demonstrate that the proposed optimized control strategy provides significant energy-saving benefits for electric scooters.

Power consumption characteristics

Power comparison under different ambient temperatures (WMTC PART 1 Mode)

In this study, a heat exchange-based cooling platform was designed to conduct experiments under various ambient temperatures (30 °C, 35 °C, 40 °C, 45 °C, and 50 °C). Regarding driving conditions, the WMTC PART 1 mode was adopted for the acquisition of water pump power consumption and coolant temperature data, followed by comparative analysis. The power consumption under the proposed optimized control strategy was then compared to that under the original control conditions to evaluate its energy-saving performance.

Figure 8 illustrates the average power consumption comparison between the original control strategy and the proposed optimized control strategy under different ambient temperature conditions (30 °C, 35 °C, 40 °C, 45 °C, and 50 °C) using the WMTC PART 1 mode. The test results show that the average power consumption of the water pump under the original control strategy ($P_{o,avg}$) was 18.35 W. In contrast, under the proposed optimized control strategy, the average power consumption ($P_{c,avg}$) was 0.94 W at 30 °C, 2.84 W at 35 °C, 3.43 W at 40 °C, 6.61 W at 45 °C, and 8.04 W at 50 °C.

Based on the calculations from Eqs. (6) and (7), the corresponding average power differences at each temperature condition were 17.41 W at 30 °C, 15.51 W at 35 °C, 14.92 W at 40 °C, 11.74 W at 45 °C, and 10.31 W at 50 °C. These results demonstrate that the proposed optimized control strategy not only reduces power consumption but also maintains thermal stability across a wide range of ambient temperatures. This indicates that adaptive control of the water pump based on coolant temperature is an effective approach for balancing thermal performance and energy efficiency. The detailed data are presented in Table 3.

Fig. 8 Comparison of average power consumption (W) between the original water pump and the proposed optimized control strategy under ambient temperatures of 30 °C, 35 °C, 40 °C, 45 °C, and 50 °C (WMTC PART 1 mode)

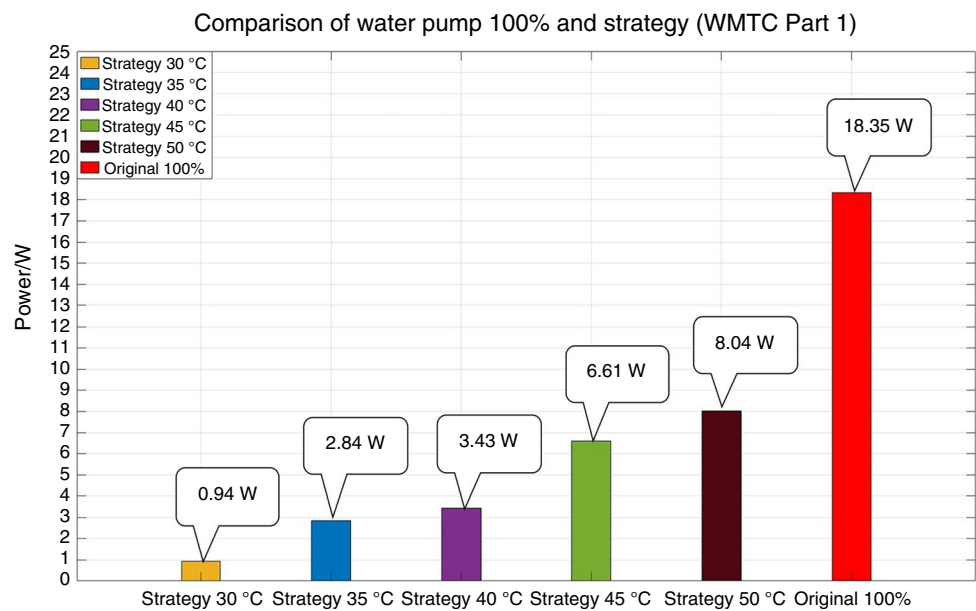


Table 3 Comparison of $P_{c,avg}$ and P_{avg} under different ambient temperatures using the proposed control strategy in WMTC PART 1 mode

Temperature/°C	Average power consumption (W) of the proposed optimized control strategy, $P_{c,avg}$ / W	Average power difference, P_{avg} / W
30	0.94	17.41
35	2.84	15.51
40	3.43	14.92
45	6.61	11.74
50	8.04	10.31

$$(P_{o,avg} - P_{c,avg}) = P_{avg} \quad (6)$$

Here, $P_{o,avg}$ denotes the average power consumption of the original water pump control strategy, while $P_{c,avg}$ represents the average power consumption under the proposed optimized control strategy. P_{avg} refers to the average power difference between the two.

The relationship among the total driving duration s (in seconds), the power per second W , and P_{avg} can be expressed by the following equation:

$$\frac{s}{P_{avg}} = W \quad (7)$$

Power consumption under different ambient temperatures at constant speed (50 km h⁻¹)

Figure 9 presents a comparison of the average power consumption between the original water pump and the proposed

optimized control strategy under the customized constant-speed mode at 50 km h⁻¹, across ambient temperatures of 30 °C, 35 °C, 40 °C, 45 °C, and 50 °C. According to the test results, the average power consumption of the original water pump ($P_{o,avg}$) was 18.26 W within the ambient temperature range of 30 °C to 50 °C. Under the proposed optimized control strategy, the average power consumption ($P_{c,avg}$) at different ambient temperatures was as follows: 1.88 W at 30 °C, 2.37 W at 35 °C, 4.73 W at 40 °C, 5.52 W at 45 °C, and 8.82 W at 50 °C. Compared with the original power consumption of 18.26 W, the corresponding power reductions, calculated using Eqs. (6) and (7), were 16.38 W at 30 °C, 15.89 W at 35 °C, 13.53 W at 40 °C, 12.74 W at 45 °C, and 9.44 W at 50 °C. These results demonstrate that the proposed optimized control strategy can significantly reduce system power consumption across all tested ambient temperatures, thereby offering substantial energy-saving benefits. Detailed numerical data are provided in Table 4.

Energy efficiency performance

Energy improvement under different ambient temperatures (WMTC PART 1 Mode)

Equations (8) and (9) represent the total power consumption under the original control strategy ($P_{o,tot}$) and the proposed optimized control strategy ($P_{t,tot}$) under various ambient temperature conditions, respectively. A programmable power supply was used to acquire power consumption data, and the tests were conducted under the WMTC PART 1 driving mode. When the ambient temperature ranged from 30 to 50 °C, the total power consumption under the original control strategy ($P_{o,tot}$) was 44,094.74 W. In contrast, the total power

Fig. 9 Comparison of average power consumption between the original water pump and the proposed optimized control strategy at ambient temperatures of 30 °C, 35 °C, 40 °C, 45 °C, and 50 °C under the 50 km h⁻¹ constant-speed mode

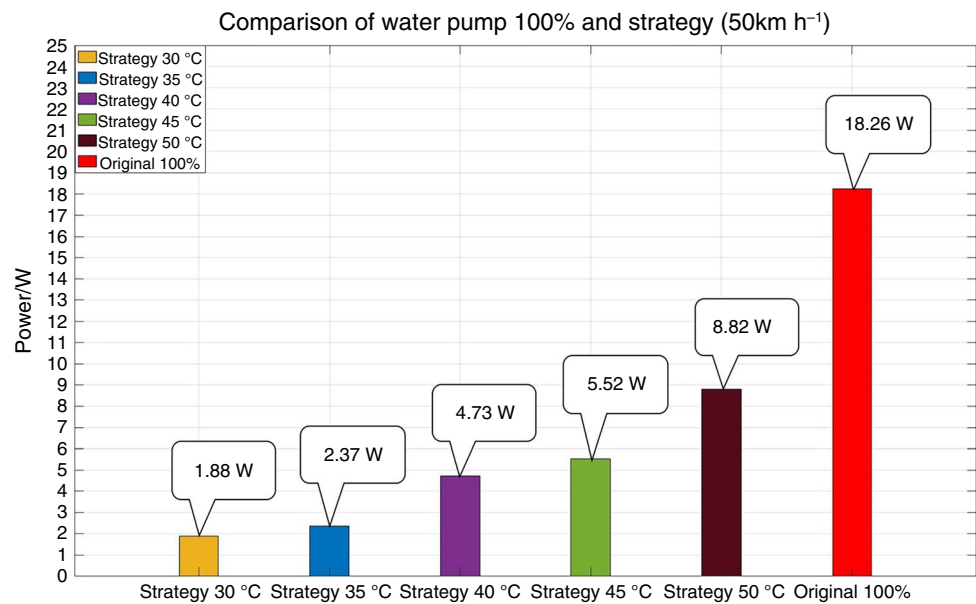


Table 4 Comparison of P_{avg} and P_{avg} under different ambient temperatures using the proposed optimized control strategy in the 50 km h⁻¹ mode

Temperature/°C	P_{avg}/W	P_{avg}/W
30	1.88	16.38
35	2.37	15.89
40	4.73	13.53
45	5.52	12.74
50	8.82	9.44

consumption under the proposed optimized control strategy (P_{tot}) was 2,266.78 W at 30 °C, 6,814.62 W at 35 °C, 8,238.98 W at 40 °C, 15,855.48 W at 45 °C, and 19,286.49 W at 50 °C. These results clearly demonstrate that the proposed control strategy significantly reduces energy consumption, highlighting its effectiveness in improving energy efficiency. The detailed data are presented in Table 5.

$$\sum_{t=1}^{t=s} P_o(t) = P_{o,tot}(t) \quad (8)$$

Table 5 Comparison of P_{tot} and E_{imp} under different ambient temperatures using the proposed optimized control strategy in WMTC PART 1 mode

Temperature/°C	Total power consumption under the proposed optimized control strategy, P_{tot}/W	Energy consumption improvement rate, $E_{imp}/\%$
30	2266.78	94.86
35	6814.62	84.55
40	8238.98	81.32
45	15855.48	64.04
50	19286.49	56.26

$$\sum_{t=1}^{t=s} P_c(t) = P_{c,tot}(t) \quad (9)$$

Here, s represents the total duration (in seconds) of the driving cycle.

The relationship among $P_{o,tot}$, P_{tot} , and the energy consumption improvement rate % (E_{imp}) can be expressed by the following equation:

$$\frac{P_{o,tot} - P_{c,tot}}{P_{o,tot}} \times 100\% = E_{imp} \quad (10)$$

Under the proposed optimized control strategy, the energy consumption improvement rate % (E_{imp}) was calculated using Eqs. (8) through (10). The results demonstrate that when the ambient temperatures were 30 °C, 35 °C, 40 °C, 45 °C, and 50 °C, the corresponding E_{imp} values were 94.86%, 84.55%, 81.32%, 64.04%, and 56.26%, respectively. These findings indicate that the proposed optimized control strategy significantly enhances energy efficiency and demonstrates notable energy-saving performance. Detailed results are presented in Table 5.

Energy efficiency under different ambient temperatures at constant speed (50 km h⁻¹)

In this study, a programmable power supply was used to measure power consumption under the customized constant-speed mode at 50 km h⁻¹. When the ambient temperature ranged from 30 to 50 °C, the total power consumption under the original control strategy ($P_{o,tot}$) was recorded as 20,089.16 W. In contrast, under the proposed optimized control strategy, the total power consumption ($P_{c,tot}$) at each temperature condition was as follows: 2,073.46 W at 30 °C, 2,604.26 W at 35 °C, 5,205.04 W at 40 °C, 6,066.63 W at 45 °C, and 9,697.24 W at 50 °C. These results clearly demonstrate the effectiveness of the proposed optimized control strategy in reducing energy consumption, as detailed in Table 6.

Furthermore, the energy improvement rate % (E_{imp}) was calculated using Eqs. (8) to (10). The calculated E_{imp} values were 89.68%, 87.04%, 74.09%, 69.80%, and 51.73% at ambient temperatures of 30 °C, 35 °C, 40 °C, 45 °C, and 50 °C, respectively. These findings confirm that the proposed

optimized control strategy significantly reduces system energy consumption across a range of ambient temperatures, demonstrating its substantial energy-saving potential. The detailed results are summarized in Table 6.

Coolant temperature behavior

Temperature variation under different ambient temperatures (WMTC PART 1 Mode)

The heat exchange-based cooling system proposed in this study (as shown in Fig. 5) was equipped with a temperature data acquisition device to record coolant temperature variations at multiple measurement points during the experiment. The measurement points were defined as follows: T1 represents the section between the coolant reservoir and the water pump; T2 is located between the traction motor and the heat exchanger; T3 lies between the heat exchanger and the MCU heater; and T4 is positioned between the MCU heater and the coolant reservoir.

Figure 10 presents the temperature–time profiles at measurement point T1 under ambient temperatures of 30 °C, 35 °C, 40 °C, 45 °C, and 50 °C, comparing the original water pump strategy with the proposed optimized control strategy (WMTC PART 1 mode). The testing duration was from 0 to 2400 s. The results demonstrate that the average temperature differences between the two strategies were 0.35 °C at 30 °C, 0.26 °C at 35 °C, 1.83 °C at 40 °C, 0.18 °C at 45 °C, and 0.33 °C at 50 °C. These findings suggest that the proposed strategy effectively suppresses temperature rise while minimizing unnecessary energy consumption, highlighting its advantage over conventional constant-speed cooling approaches.

Fig. 10 Temperature–time curves at measurement point T1 under ambient temperatures of 30 °C, 35 °C, 40 °C, 45 °C, and 50 °C, comparing the original water pump with the proposed optimized control strategy (WMTC PART 1 mode)

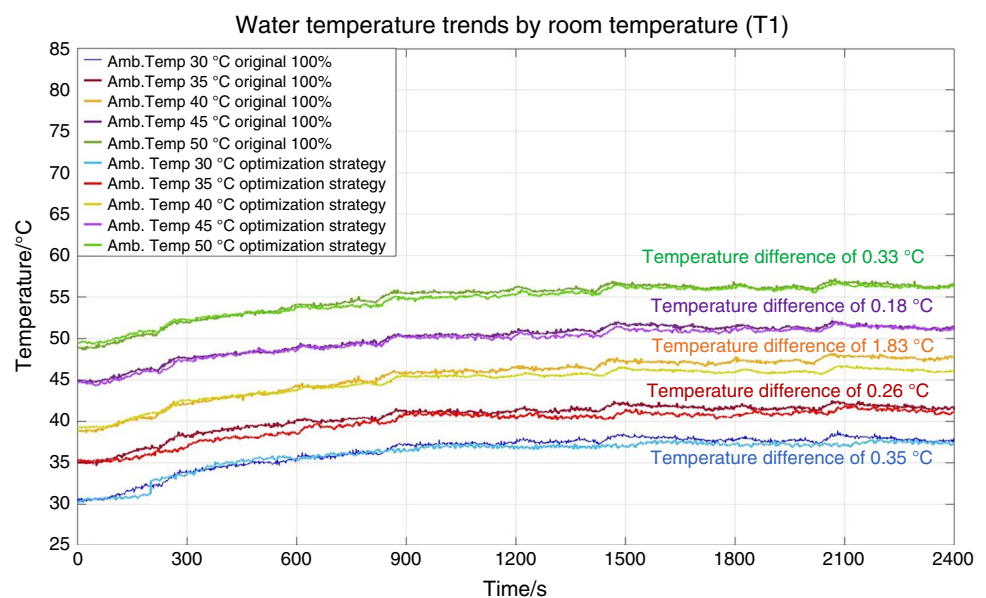


Figure 11 illustrates the temperature–time curves at measurement point T2 under the same ambient temperature conditions. The test duration was also from 0 to 2400 s. The results indicate average temperature differences of 0.1 °C at 30 °C, 0.73 °C at 35 °C, 1.1 °C at 40 °C, and 0.1 °C at both 45 °C and 50 °C. Overall, the proposed optimized control strategy demonstrates effective thermal suppression at point T2, positively impacting system performance.

Figure 12 shows the temperature–time curves at measurement point T3 for ambient temperatures of 30 °C, 35 °C, 40 °C, 45 °C, and 50 °C, with a test duration of 0–2400 s. The analysis reveals average temperature differences of 1.05 °C at 30 °C, 1.00 °C at 35 °C, 1.38 °C at 40 °C, 0.02 °C

at 45 °C, and 0.37 °C at 50 °C. These results indicate that the proposed optimized control strategy provides effective thermal suppression under both high and low temperature conditions at point T3, contributing to stable system operation and improved overall performance.

Figure 13 presents the temperature–time profiles at measurement point T4 under ambient temperatures of 30 °C, 35 °C, 40 °C, 45 °C, and 50 °C, with the test duration ranging from 0 to 2400 s. The experimental results show that the average temperature differences between the two strategies were 0.9 °C at 30 °C, 0.74 °C at 35 °C, 2.06 °C at 40 °C, 0.39 °C at 45 °C, and 0.21 °C at 50 °C. Overall, the consistent results from Figs. 10 through 13 confirm that

Fig. 11 Temperature–time curves at measurement point T2 under ambient temperatures of 30 °C, 35 °C, 40 °C, 45 °C, and 50 °C, comparing the original water pump with the proposed optimized control strategy (WMTC PART 1 mode)

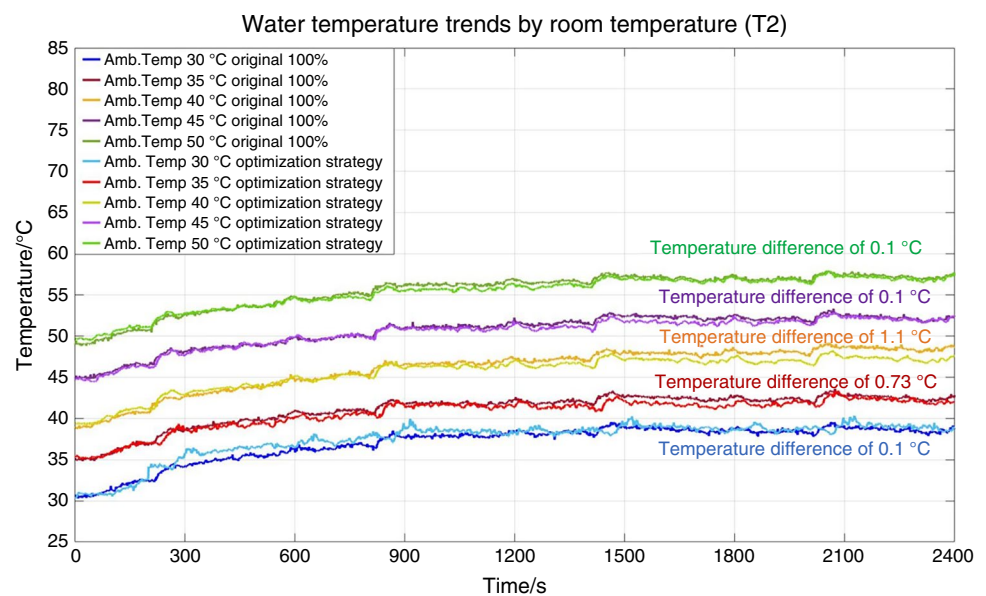


Fig. 12 Temperature–time curves at measurement point T3 under ambient temperatures of 30 °C, 35 °C, 40 °C, 45 °C, and 50 °C, comparing the original water pump with the proposed optimized control strategy (WMTC PART 1 mode)

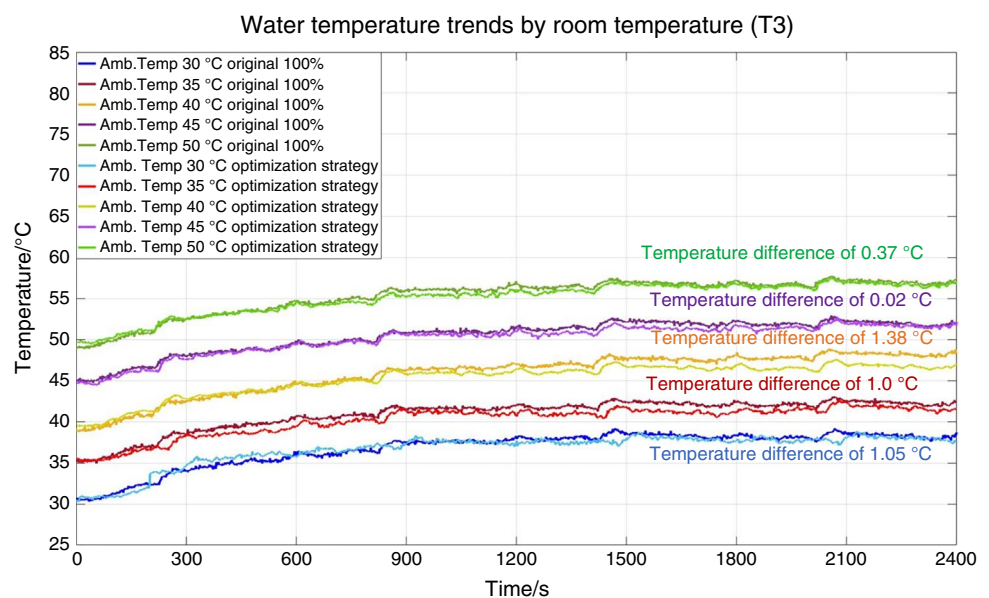
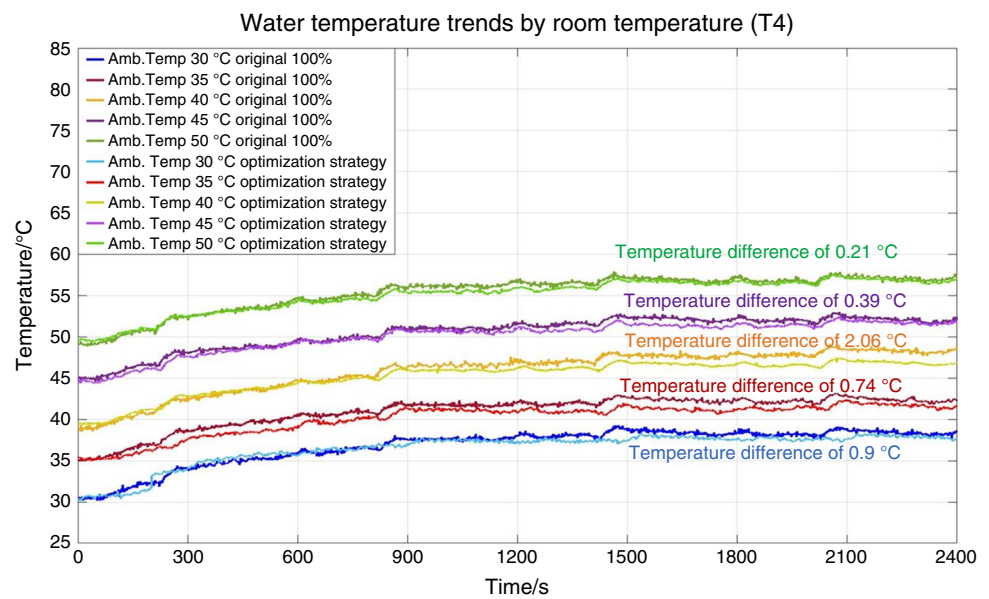


Fig. 13 Temperature–time curves at measurement point T4 under ambient temperatures of 30 °C, 35 °C, 40 °C, 45 °C, and 50 °C, comparing the original water pump with the proposed optimized control strategy (WMTC PART 1 mode)



the proposed optimized control strategy exhibits stable and effective temperature suppression, thereby enhancing the overall operational performance of the system.

Table 7 presents the average temperature differences between the proposed optimized control strategy and the original water pump at various measurement points (T1, T2, T3, T4) under different ambient temperature conditions in WMTC PART 1 mode. The experimental results indicate that the proposed optimized control strategy effectively suppresses system temperature rise, with the best performance observed at 40 °C, followed by 35 °C, 30 °C, 50 °C, and 45 °C, in descending order.

Coolant temperature under different ambient temperatures at constant speed (50 km h⁻¹)

The heat exchange cooling system proposed in this study (as shown in Fig. 5) was evaluated using a temperature data acquisition device during the experiment to record coolant temperatures at various measurement points. The definitions of each measurement point are as follows: T1 refers to the section between the coolant reservoir and the

water pump; T2 is located between the traction motor and the heat exchanger; T3 corresponds to the section between the heat exchanger and the MCU heater; and T4 is situated between the MCU heater and the coolant reservoir.

Figure 14 presents the temperature–time curves at measurement point T1 under ambient temperatures of 30 °C, 35 °C, 40 °C, 45 °C, and 50 °C, comparing the original water pump with the proposed optimized control strategy in the 50 km h⁻¹ mode. The testing duration ranged from 0 to 1100 s. The results demonstrate that the average temperature differences between the two configurations were 2.53 °C at 30 °C, 2.03 °C at 35 °C, 1.11 °C at 40 °C, 0.61 °C at 45 °C, and 1.93 °C at 50 °C.

Figure 15 shows the temperature–time curves at measurement point T2 under the same ambient temperature conditions, also in the 50 km h⁻¹ mode and within the same test duration (0–1100 s). The corresponding average temperature differences were 1.56 °C at 30 °C, 1.11 °C at 35 °C, 0.83 °C at 40 °C, 0.34 °C at 45 °C, and 1.89 °C at 50 °C. These results demonstrate that the proposed optimized control strategy effectively suppresses temperature

Table 7 Average temperature differences between the proposed optimized control strategy and the original water pump at measurement points (T1, T2, T3, T4) under different ambient temperatures in WMTC PART 1 mode

Temperature/°C	Measurement point T1/°C	Measurement point T2/°C	Measurement point T3/°C	Measurement point T4/°C	Average value of measurement points T1 to T4/°C
30	0.35	0.1	1.05	0.9	0.6
35	0.26	0.73	1.00	0.74	0.68
40	1.83	1.1	1.38	2.06	1.59
45	0.18	0.1	0.02	0.39	0.17
50	0.33	0.1	0.37	0.21	0.25

Fig. 14 Temperature–time curves at measurement point T1 under ambient temperatures of 30 °C, 35 °C, 40 °C, 45 °C, and 50 °C, comparing the original water pump with the proposed optimized control strategy in the 50 km h⁻¹ mode

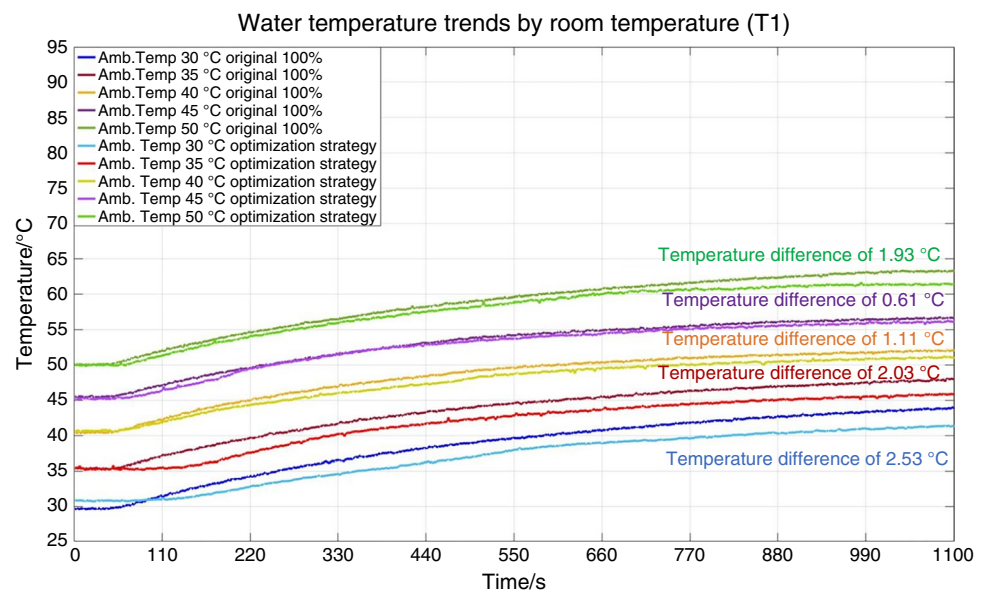
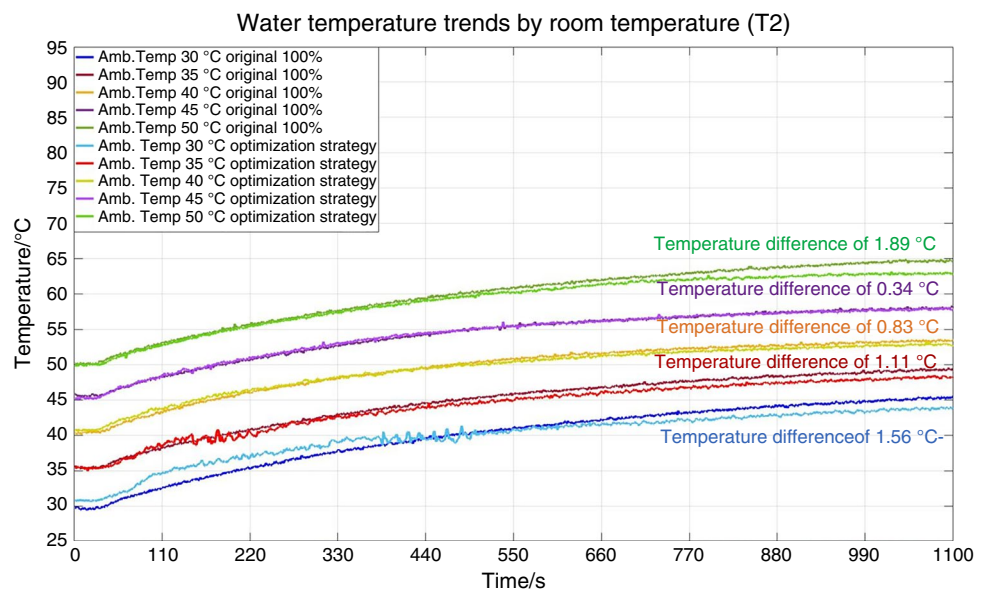


Fig. 15 Temperature–time curves at measurement point T2 under ambient temperatures of 30 °C, 35 °C, 40 °C, 45 °C, and 50 °C, comparing the original water pump with the proposed optimized control strategy in the km h⁻¹ mode



rise at both T1 and T2, contributing to improved overall system performance.

Figure 16 presents the temperature–time curves at measurement point T3 under identical test conditions. The average temperature differences were recorded as 1.98 °C at 30 °C, 1.93 °C at 35 °C, 1.02 °C at 40 °C, 0.65 °C at 45 °C, and 2.30 °C at 50 °C. Figure 17 shows the temperature–time curves at measurement point T4, with the same ambient temperature conditions and testing duration. The corresponding average temperature differences were 2.37 °C at 30 °C, 1.82 °C at 35 °C, 1.08 °C at 40 °C, 0.66 °C at 45 °C, and 2.18 °C at 50 °C. Taken together, the results from Figs. 14 through 17 confirm that the proposed optimized control

strategy consistently exhibits stable and effective suppression of temperature rise across all measurement points and ambient temperature conditions. This contributes significantly to enhanced thermal management and overall performance of the cooling system.

Table 8 presents the average temperature differences between the original water pump and the proposed optimized control strategy at various measurement points (T1, T2, T3, and T4) under different ambient temperature conditions in the 50 km h⁻¹ mode. The results indicate that the proposed optimized control strategy effectively mitigates system temperature rise, with the best performance observed at 30 °C, followed by 50 °C, 35 °C, 40 °C, and 45 °C, in that order.

Fig. 16 Temperature–time curves at measurement point T3 under ambient temperatures of 30 °C, 35 °C, 40 °C, 45 °C, and 50 °C, comparing the original water pump with the proposed optimized control strategy in the 50 km h⁻¹ mode

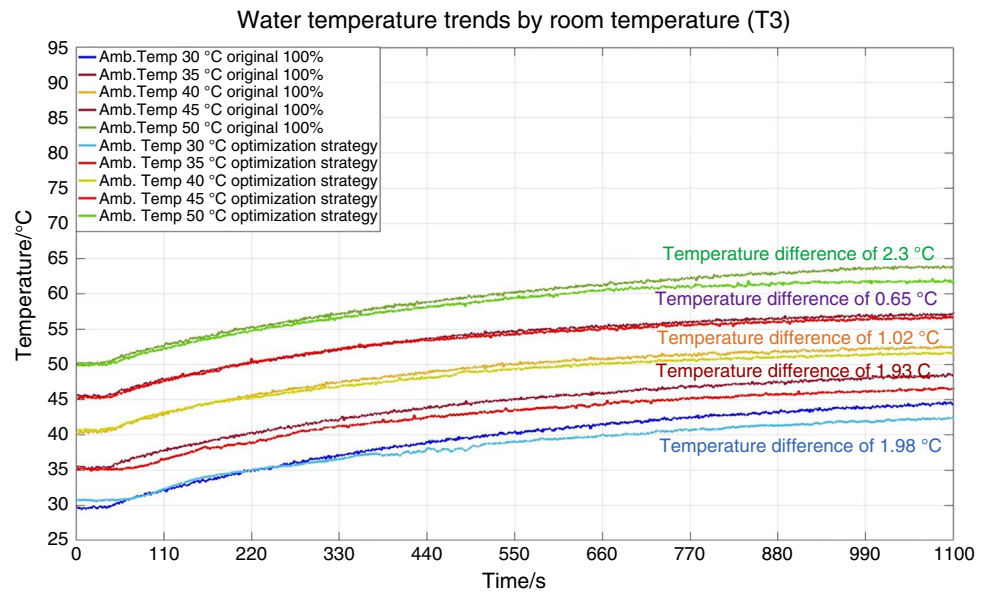


Fig. 17 Temperature–time curves at measurement point T4 under ambient temperatures of 30 °C, 35 °C, 40 °C, 45 °C, and 50 °C, comparing the original water pump with the proposed optimized control strategy in the km h⁻¹ mode

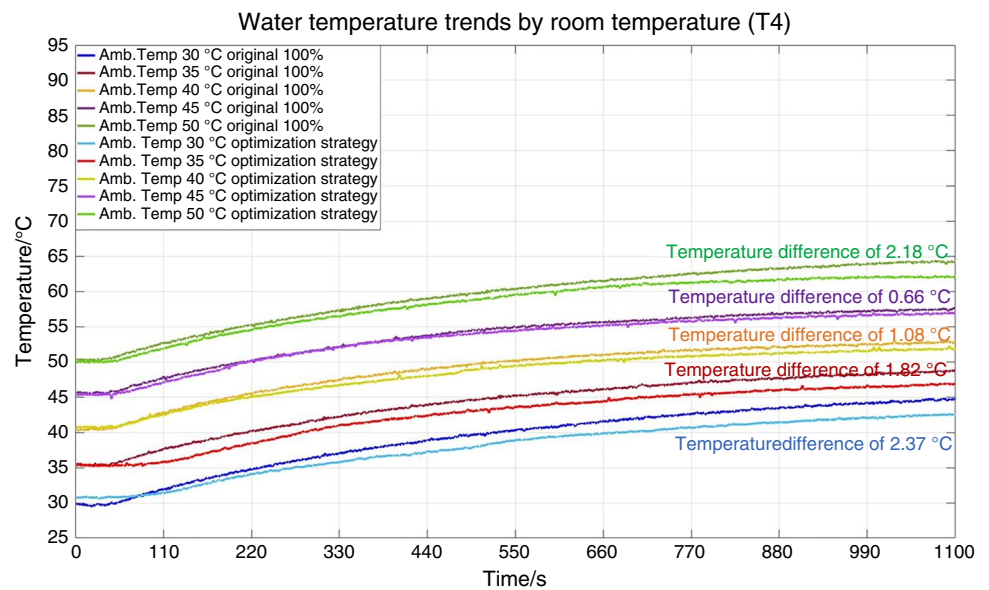


Table 8 Average temperature differences between the proposed optimized control strategy and the original water pump at measurement points (T1, T2, T3, T4) under different ambient temperatures in 50 km h⁻¹ mode

Temperature/°C	Measurement point T1/°C	Measurement point T2/°C	Measurement point T3/°C	Measurement point T4/°C	Average value of measurement points T1 to T4/°C
30	2.53	1.56	1.98	2.37	2.11
35	2.03	1.11	1.93	1.82	1.72
40	1.11	0.83	1.02	1.08	1.01
45	0.61	0.34	0.65	0.66	0.57
50	1.93	1.89	2.30	2.18	2.08

Conclusions

This study established a thermal exchange and cooling simulation platform compatible with the 6.4 kW Gogoro 2 electric scooter to investigate the effects of rising ambient temperature on vehicle performance. Experimental observations revealed that as the ambient temperature increases, the coolant temperature also rises, thereby affecting the overall performance of the electric scooter. This phenomenon is primarily attributed to the thermal management system's inability to dissipate heat effectively and in real time, leading to elevated temperatures in the motor and other critical components, which in turn reduces the vehicle's operational efficiency and stability. To mitigate the adverse effects of high ambient temperatures on electric scooter performance, this study proposes an optimized control strategy designed to regulate the coolant temperature effectively and prevent thermal runaway during operation in hot environments. The proposed strategy not only adjusts the cooling system dynamically in response to rising ambient temperatures but also utilizes real-time temperature data from the vehicle to control the water pump operation adaptively. When the coolant temperature exceeds 30 °C, the proposed optimized control strategy automatically activates the cooling system to maintain the coolant temperature within a safe range, thereby preventing overheating. By adjusting the water pump duty cycle according to the temperature range, the proposed strategy successfully suppresses system temperature rise while also achieving favorable energy-saving performance.

In the WMTC PART 1 mode test, when the ambient temperature ranged from 30 to 50 °C, the average power consumption of the original water pump was measured at 18.35 W. After applying the optimized control strategy proposed in this study, the average power reduction and energy improvement rate % (E_{imp}) under different ambient temperatures were as follows: 17.41 W and 94.86% at 30 °C; 15.51 W and 84.55% at 35 °C; 14.92 W and 81.32% at 40 °C; 11.74 W and 64.04% at 45 °C; and 10.31 W and 56.26% at 50 °C. These results clearly indicate that the proposed optimized control strategy provides a significant energy-saving effect and effectively reduces system power consumption.

In the customized constant-speed 50 km h⁻¹ mode test, when the ambient temperature was between 30 and 50 °C, the average power consumption of the original water pump was 18.26 W. With the proposed optimized control strategy applied, the corresponding average power reductions and E_{imp} values were: 16.38 W and 89.68% at 30 °C; 15.89 W and 87.04% at 35 °C; 13.53 W and 74.09% at 40 °C; 12.74 W and 69.80% at 45 °C; and 9.44 W and 51.73% at 50 °C. These findings confirm that the proposed strategy

demonstrates stable and substantial energy-saving performance under varying operating modes and ambient temperature conditions.

Through the regulation mechanism of the proposed optimized control strategy, the cooling system can operate efficiently under various temperature conditions, thereby protecting critical components of the EV and reducing the risk of thermal runaway due to overheating. In addition to effectively managing coolant temperature, the proposed optimized control strategy also contributes to a significant improvement in the vehicle's driving range. Research findings indicate that continuous operation of the water pump leads to additional energy consumption. However, the proposed optimized control strategy developed in this study dynamically adjusts the water pump's operating cycle based on real-time demand, effectively reducing energy consumption and consequently extending the EV's driving range. Compared with conventional fixed-duty cooling strategies, the proposed approach introduces a simple yet effective rule-based control mechanism that adapts to real-time thermal conditions, thereby improving both system efficiency and operational reliability.

The proposed control strategy enhances thermal management performance while effectively reducing the energy consumption of the cooling system, thereby improving overall energy efficiency. Under certain operating conditions, it also demonstrates the potential to extend the driving range of electric scooters. From an application perspective, the proposed thermal management control strategy can be directly implemented in electric scooters and other small electric vehicles equipped with liquid-cooling systems. This temperature-dependent control logic enables real-time regulation of the water pump without relying on complex optimization algorithms or additional computational resources, offering strong practicality and ease of implementation. Furthermore, in industrial applications, this approach can be integrated into existing vehicle control units (VCUs) or embedded systems as a cost-effective solution for improving thermal regulation and reducing auxiliary power consumption. This is particularly beneficial for electric vehicles operating in high-temperature environments, where effective thermal management is critical for maintaining system reliability and performance.

In addition, the proposed method can further extend the driving range by minimizing unnecessary power consumption of the cooling system. However, although the results indicate an improvement in driving range, the extent of this increase remains limited under the current test conditions. Therefore, this study primarily focuses on the directly measurable improvements in energy consumption and thermal regulation performance. Overall, the proposed approach shows strong potential and practical value for future applications in commercial electric vehicle platforms.

Future work will focus on the optimization of geometric design and flow control parameters of the cooling system. In particular, further investigations will be conducted to analyze the effects of coolant flow distribution and system configuration on thermal performance. In addition, integrating detailed flow field analysis and system-level modeling will help to further improve the effectiveness and applicability of the proposed control strategy.

Author contribution Chun-Hsin Chang contributed to conceptualization, validation, writing—original draft, software, review & editing, methodology, and supervision. Guan-Xun Wang and Hwa-Dong Liu helped in writing—original draft.

Funding No funding was received for this manuscript.

Data availability Not applicable.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical approval Not applicable.

Consent to participate Not applicable.

Consent for publications Not applicable.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

1. Khoshnevisan A, Changizian S, Raeesi M, Ahmadi P, Javani N. Thermal analysis of thermo-electric generator systems in hybrid electric vehicles under different operating conditions. *J Therm Anal Calorim*. 2023. <https://doi.org/10.1007/s10973-023-12349-0>.
2. Pavai PS, Chandrakanth B, Madivada H, Mallisetty PK, Sharma TK. Energy and exergy analysis of ternary nanofluid for electric vehicle coolant through invasive weed optimisation algorithm—a numerical study. *J Therm Anal Calorim*. 2024. <https://doi.org/10.1007/s10973-024-13698-0>.
3. Nadimuthu LPR, Victor K, Karthikeyan PN, Premkumar IJI, Naavaneethakrishnan G, Palanisamy R, et al. Green agro storage and electric vehicle integrated nano grid for rural livelihood improvement: a detailed review and case analysis. *J Therm Anal Calorim*. 2024. <https://doi.org/10.1007/s10973-024-13515-8>.
4. Pavai PS, Chandrakanth B, Mallisetty PK, Sharma TK, Suresh B, Ansari MM. Experimental assessment of a hybrid radiator for pure electric vehicle traction system: a 4E approach (energy, exergy, environment and economy). *J Therm Anal Calorim*. 2025. <https://doi.org/10.1007/s10973-025-14086-y>.
5. Shewalkar AG, Dhoble AS, Thawkar VP. Review on cooling techniques and analysis methods of an electric vehicle motor. *J Therm Anal Calorim*. 2024. <https://doi.org/10.1007/s10973-024-13091-x>.
6. Hamwi H, Rushby T, Mahdy M, Bahaj AS. Effects of high ambient temperature on electric vehicle efficiency and range: case study of Kuwait. *Energies*. 2022. <https://doi.org/10.3390/en15093178>.
7. Yuksel T, Michalek JJ. Effects of regional temperature on electric vehicle efficiency, range, and emissions in the United States. *Environ Sci Technol*. 2015. <https://doi.org/10.1021/es505621s>.
8. Wang X, et al. A critical review on thermal management technologies for motors in electric cars. *Appl Therm Eng*. 2022. <https://doi.org/10.1016/j.applthermaleng.2021.117758>.
9. Fan J, et al. Thermal analysis of permanent magnet motor for the electric vehicle application considering driving duty cycle. *IEEE Trans Magn*. 2010. <https://doi.org/10.1109/TMAG.2010.2042043>.
10. Tikadar A, et al. Comparison of electro-thermal performance of advanced cooling techniques for electric vehicle motors. *Appl Therm Eng*. 2021. <https://doi.org/10.1016/j.applthermaleng.2020.116182>.
11. Raj JAPS, et al. Thermal management strategies and power ratings of electric vehicle motors. *Renew Sustain Energy Rev*. 2024. <https://doi.org/10.1016/j.rser.2023.113874>.
12. Li C, Cheng S, Li J, Li H, et al. Heat dissipation evaluation and optimization of air cooling induction motor used for mini electric vehicle. *Proceedings of the 20th International Conference on Electrical Machines and Systems (ICEMS)*. 2017. <https://doi.org/10.1109/ICEMS.2017.8056437>.
13. Tian Z, et al. Investigation on an integrated thermal management system with battery cooling and motor waste heat recovery for electric vehicle. *Appl Therm Eng*. 2018. <https://doi.org/10.1016/j.applthermaleng.2018.02.093>.
14. Szumska EM, Jurecki RS. Parameters influencing on electric vehicle range. *Energies*. 2021. <https://doi.org/10.3390/en14164821>.
15. Ali HM. Thermal management systems for batteries in electric vehicles: a recent review. *Energy Rep*. 2023. <https://doi.org/10.1016/j.egy.2023.04.359>.
16. Molaieimanesh GR, et al. Experimental analysis of commercial LiFePO4 battery life span used in electric vehicle under extremely cold and hot thermal conditions. *J Therm Anal Calorim*. 2021. <https://doi.org/10.1007/s10973-020-09272-z>.
17. Wang A, Li Q, Cao F, Hu D. Driving range evaluation of electric vehicle with transcritical CO2 thermal management system under different battery temperature controls. *J Clean Prod*. 2024. <https://doi.org/10.1016/j.jclepro.2023.140208>.
18. Ibrahim A, Guo J, Wang Y, Zheng Y, Lei B, et al. Performance of serpentine channel based Li-ion battery thermal management system: an experimental investigation. *Int J Energy Res*. 2020. <https://doi.org/10.1002/er.5599>.
19. Ouyang D, Weng J, Hu J, Liu J, Chen M, Huang Q, et al. Effect of high temperature circumstance on lithium-ion battery and the application of phase change material. *J Electrochem Soc*. 2019. <https://doi.org/10.1149/2.0441904jes>.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.